Deriving the Output Response qst from an Elasticity Assumption

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What question are we answering?

When a carbon policy changes a firm's unit cost of production, we need a simple way to translate that cost shock into a change in firm output. This derivation builds a reduced-form rule that maps unit cost changes into post-policy output (qst) using a elasticity assumption.

- Input: unit cost change per unit output (del_u) produced by technology adoption and carbon trading.
- Output: new firm output level under the policy. (qst).
- Mechanism: a constant elasticity relationship between output and a dimensionless competitiveness index.

Key variables and units

Symbol	Meaning
q0	Baseline output (units of product).
qst	Output after the policy (units of product).
py	Unit product price ($/unit). In this model we assume baseline zero profit, so py is also the baseline unit production cost.
del_u	Change in unit cost caused by the policy response ($/unit).
δ	Proportional unit-cost shock, δ = del_u / py (dimensionless).
M	Competitiveness (or remaining-price) index, M = (py - del_u) / py = 1 - δ (dimensionless).
ε	Elasticity parameter controlling how sensitively output responds to changes in M (dimensionless).

Step-by-step derivation

Step 1: Build a dimensionless driver for output response.

Elasticities are defined for proportional (percent) changes, so we first convert the unit cost change del_u (in $/unit) into a unitless index. With baseline zero profit (py equals baseline unit cost), the proportional unit-cost shock is:

$$\delta = \mathrm{del\_u} / \mathrm{py}$$

We then define a competitiveness (or remaining-price) index that falls when unit costs rise:

$$M = (py - del\_u) / py = 1 - \delta$$

Interpretation: M = 1 means no cost shock; M < 1 means the policy raises unit cost as a share of price; M > 1 means unit costs fall (a net benefit).

Step 2: Impose a constant-elasticity relationship between output and M.

We assume that output responds smoothly to the competitiveness index with constant elasticity ε. The elasticity definition is:

$$\varepsilon = d\ln(q)/d\ln(M)$$

This says: a 1% change in M produces an ε% change in output q, holding other factors constant.

Step 3: Convert the elasticity definition into a differential equation.

Starting from $\varepsilon = d\ln(q) / d\ln(M)$, multiply both sides by $d\ln(M)$:

$$d\ln(q) = \varepsilon\, d\ln(M)$$

Using $d\ln(q) = dq/q$ and $d\ln(M) = dM/M$, this becomes:

$$dq/q = \varepsilon\,(dM/M)$$

Step 4: Integrate from baseline to the post-policy outcome.

Integrate both sides from the baseline state (q0, M0) to the post-policy state (qst, M):

$$\int_{q0}^{qst} dq/q = \varepsilon \int_{M0}^{M} dM/M$$

The integrals evaluate to logarithms:

$$\ln(qst/q0) = \varepsilon\ln(M/M0)$$

Exponentiating both sides yields the power-law response:

$$qst = q0\,(M/M0)^{\varepsilon}$$

Step 5: Apply the baseline normalization and substitute M.

At the baseline (no policy cost shock), $del\_u = 0$, so δ = 0 and $M0 = 1$. Therefore

$$qst = q0(M)^{\varepsilon}$$

Substituting

$$M = 1 - del\_u/py$$

gives:

$$\text{qst} = \text{q0}\,(1 - \text{del\_u/py})^{\varepsilon}$$

Notes on interpretation

This output response is reduced-form: it is designed to capture the direction and magnitude of output changes when unit costs change, without explicitly solving a full product-market equilibrium. The elasticity ε is a calibration parameter: larger ε implies stronger output response to the same proportional cost shock.